

20SLG1

SomeonecoolLovesMaths

June 6, 2026

Problem

Let ABC be an isosceles triangle with $BC = CA$, and let D be a point inside side AB such that $AD < DB$. Let P and Q be two points inside sides BC and CA , respectively, such that $\angle DPB = \angle DQA = 90^\circ$. Let the perpendicular bisector of PQ meet line segment CQ at E , and let the circumcircles of triangles ABC and CPQ meet again at point F , different from C . Suppose that P, E, F are collinear. Prove that $\angle ACB = 90^\circ$.

Solution. Let M be the midpoint of AB . As a result, $CM \perp AB$. $\therefore \angle CQD = -\angle CPD = -\angle CMD$, $CPMDQF$ are concyclic.

Let X be the midpoint of PQ . $\angle EQP = \angle CFP = \angle EPQ = \angle FCQ$. So EX is the perpendicular bisector of FC as well. If we are able to show that EX intersects M then we are done as then M will become the intersection of 2 perpendicular bisectors of non-parallel chords, which is the circumcenter, after which we are obviously done.

Since $\angle QCM = \angle PCM$, $PM = MQ$. As a result $\triangle MXQ \cong \triangle MXP$. This of course implies that E, X, M are collinear.